

Technical Review & Build Verdict

Financial Forecasting by Multi-AI Agent System — B.Tech Capstone

Reviewer role: Finance-AI research review **Review date:** 23 July 2026 **Scope:** BTP_All_Phases_Work_Breakdown.pdf (5 phases), existing repository state, and the underlying research premise **Method:** Every factual claim below was checked against a primary source or reproduced on this machine. Claims that could **not** be verified are marked [!] **UNVERIFIED** and listed separately in §9 — they are not presented as findings.

1. Verdict

[OK] BUILD — but do not build the plan as written.

The architecture is sound and the engineering instincts are good. The project does **not** need a restart. However, **as currently planned, the project would pass its demo and fail its evaluation chapter.** Three things must be fixed *before* Phase 2 code is written, because retrofitting them later means touching every tool and every agent:

#	Must fix before building	Why it cannot wait
1	Add <code>as_of</code> to every data function	Retrofitting means rewriting all 6 tools <i>and</i> re-running all experiments
2	Move evaluation before the build; define a post-cutoff test window	A within-training-window backtest is invalid; discovering this in Phase 5 leaves no time
3	Reposition the novelty claim against TradingAgents	Affects the report's framing, the research question, and what you measure

Everything else in this review is fix-as-you-go.
Estimated cost of these three fixes: ~1 day of planning. Cost of not fixing them: the results chapter.

2. What Was Verified, and How

#	Claim under test	Method	Result
1	<code>pandas-ta-classic</code> import name	PyPI package page	[X] Bug confirmed — imports as <code>pandas_ta_classic</code>
2	Project venv state	Direct <code>site-packages</code> inspection	[X] 0 packages installed
3	<code>tests/test_env.py</code> runs	Executed on this machine	[X] All 5 checks fail
4	yfinance rate ceiling " <code>~2,000/hr</code> "	Maintainer issue tracker	[X] False — <code>~360/hr</code> ; 429s reported from <code>~950</code>
5	LangGraph parallel state writes	LangChain error docs	[OK] Confirmed — needs <code>Annotated</code> reducer
6	Gemini nested-Pydantic reliability	google-genai + pydantic-ai issue trackers	[!] New risk found (see §4.4)
7	Gemini free-tier RPM/RPD	Official rate-limit docs	[!] Not publishable — Google no longer lists numbers

#	Claim under test	Method	Result
8	Lookahead bias is real & measurable	arXiv:2512.23847	[OK] Confirmed
9	Architecture already published	arXiv:2412.20138 + GitHub	[OK] Confirmed
10	NewsAPI free-tier viability	Vendor pricing pages	[OK] Confirmed unusable for backtest
11	GDELT global + free	gdeltproject.org	[OK] Confirmed — 100+ languages, 1979→present
12	yfinance NSE/BSE coverage	Live probe 2026-08-06	[OK] Confirmed — RELIANCE.NS , INR, Asia/Kolkata, from 1996
13	FRED needs its own key	FRED API docs	[OK] Confirmed — separate registration
14	Recharts candlestick support	Library docs	[OK] Confirmed — no native candlestick
15	Stooq keyless CSV fallback	Live probe 2026-08-06	[X] Refuted — now a JS proof-of-work page. Fallback replaced (§6.1)
16	NSE bhavcopy archive keyless	Live probe 2026-08-06	[OK] Confirmed — 200 OK w/ browser UA; 404 on holidays
17	FRED India macro series	Live probe 2026-08-06	[!] Partial — FX/10Y/call-rate live; CPI, IIP, discount rate stale or dead (§6.3)
18	GDELT rate-limit behaviour	Live probe 2026-08-06	[!] Returns the limit message as HTTP 200 body — cacheable as fake data (§6.2)

3. Blocking Bugs — Reproduced on This Machine

These are not opinions. They were executed.

3.1 The environment is empty

```
venv/Lib/site-packages → 0 non-pip packages
```

The virtualenv exists; `pip install` never completed. **Phase 1 Task 1.2 is not done**, despite the folder structure suggesting otherwise.

3.2 Every import fails

```
[X] yfinance [X] pandas_ta [X] google-genai [X] langgraph
[X] config error: No module named 'backend'
```

3.3 `import pandas_ta` will fail even after installing correctly [!] Highest-value catch

`tests/test_env.py:14` reads `import pandas_ta` as `ta`.

But `pandas-ta-classic v0.6.52` imports as `pandas_ta_classic` — verified on PyPI:

```
import pandas_ta_classic as ta # [OK] correct
import pandas_ta as ta # [X] ModuleNotFoundError
```

Without this fix, the test reports failure forever and you conclude the install is broken when it isn't. Also note: **requires Python ≥ 3.10**.

Fix:

```
import pandas_ta_classic as ta
```

3.4 No module named 'backend'

Two causes: `backend/` has no `__init__.py`, and running `python tests/test_env.py` puts `tests/` on `sys.path`, not the project root.

Fix: add `backend/__init__.py` and run as a module from the project root:

```
python -m tests.test_env
```

4. Design & Correctness Problems

4.1 The data layer leaks the future by construction

Phase 2 specifies:

```
get_price_history(ticker, period)
get_fundamentals(ticker)
get_news(ticker)
get_macro_data()
```

None accept a date. They always return *now*. Phase 5 then asks you to "pretend today is 3 months in the past."

The result: your sentiment agent reads **this week's** headlines — which already contain the outcome — while claiming to forecast from three months ago. That is not forecasting; it is hindsight. A panel finds this in under a minute.

Fix — build into Phase 2, not Phase 5:

```
get_price_history(ticker, start, end, as_of)
get_fundamentals(ticker, as_of)           # filter to filings published <= as_of
get_news(ticker, as_of)                  # filter to published_date <= as_of
get_macro_data(as_of)                   # FRED series respecting vintage
```

Enforce the cutoff **in the data layer**, never by prompting the model.

4.2 The backtest is invalid even after 4.1 is fixed Most important finding

This is the point most reviewers stop one step short of.

Fixing `as_of` removes leakage from the **pipeline**. It does nothing about leakage inside the **model's weights**. Gemini has already read what happened to these stocks. Per [arXiv:2512.23847](#), *Lookahead Propensity* is materially positive throughout the training window and **collapses to ≈ 0 immediately after the training cutoff**.

So a point-in-time-correct backtest over a within-training-window period is still meaningless.

Fix (both are required, not either/or):

1. `as_of` plumbing everywhere (§4.1), **and**
2. Test window placed **strictly after the model's training cutoff**, confirmed with a recall probe: ask the model about outcomes it should not know and verify it cannot recall them. Publish that probe as a figure — it is a genuine methodological contribution.

4.3 Cache design defeats its own purpose

The plan says: *cache is fresh if* `< 15 min old`.

For historical evaluation data keyed by an `as-of` date, **the value never changes**. A 15-minute TTL forces constant re-fetching during ablation sweeps — causing exactly the 429s the cache exists to prevent.

Fix — two cache classes:

Class	Key	TTL
Historical / point-in-time	(ticker, field, as_of)	Permanent
Live quote (demo only)	(ticker, field)	15 min

Cache **LLM responses** too, keyed by a prompt hash. Debugging re-runs must not consume fresh quota.

4.4 [!] Gemini structured output will fight your nested schemas — new risk, not previously flagged

Task 3.1 defines `FinalForecast` with nested sub-objects. Verified issues in Google's own tracker:

- **Gemini rejects Pydantic schemas that use `Field(default=...)`** (googleapis/python-genai #699). Your schemas will almost certainly use defaults.
- **Nested Pydantic models can be misread as tool calls**, producing `Unknown tool name: 'NestedModel'` or `MALFORMED_FUNCTION_CALL`.
- `propertyOrdering` improves output quality by up to ~15% on complex schemas — and models do *not* respect your declared field order without it.

Fix:

1. Test `FinalForecast` — your most complex schema — against the API in **Phase 1**, not at integration time.
2. Avoid `Field(default=...)` in response schemas; use `Optional[...]` and post-process.
3. Set `propertyOrdering` explicitly.
4. Keep nesting shallow — prefer flat schemas with prefixed field names over deep object trees.
5. Count parse failures as a **reported metric**, not a silent retry.

4.5 Parallel agents will crash LangGraph without a reducer

Task 3.5 fans out 4 specialists in parallel. If two nodes write the same state key in one step, LangGraph raises `INVALID_CONCURRENT_GRAPH_UPDATE` — *"Can receive only one value per step."*

Fix — either:

```
# (a) distinct keys per agent ← simplest, recommended
class AgentState(TypedDict):
    technical_report: TechnicalReport | None
    fundamental_report: FundamentalReport | None
    sentiment_report: SentimentReport | None
    macro_report: MacroReport | None

# (b) or a reducer, if they share a key
from typing import Annotated
import operator
reports: Annotated[list, operator.add]
```

4.6 Smaller confirmed defects

Issue	Impact	Fix
SMA-200 on a 1M window	Silent NaN	Always fetch ≥ 2 years regardless of chart range
Bhavcopy $\neq$ yfinance schema	Fallback breaks during the outage it exists for	Normalise columns and adjustment basis at the boundary (bhavcopy is unadjusted); record the source on every frame; unit-test the fallback path
yfinance nulls on .NS fundamentals	Agent crashes or invents a number	Explicit null handling; agents must say "unavailable", never guess. Log the null rate per field — it is a results table, and it explains any large-cap vs mid-cap accuracy gap

Issue	Impact	Fix
No news dedup	Same story double-counts, skewing polarity	Dedup by URL + title similarity
No version pinning	LangGraph breaking changes	<code>pip freeze > requirements.txt</code> after first good install
FRED key not provisioned	Blocks Task 2.6 mid-Phase 2	Register in Phase 1 alongside Gemini
Recharts has no candlestick	Fighting the library	Use TradingView Lightweight Charts
POST vs WebSocket unowned	30–90 s request times out on proxies	WebSocket owns the live flow; POST only for cached results
Non-deterministic runs	Reported numbers shift before demo day	temperature=0 on synthesis; cache the actual scored run

5. The Research Problem

5.1 The architecture is already published

TradingAgents: Multi-Agents LLM Financial Trading Framework — arXiv:2412.20138

Technical / fundamental / sentiment analysts, **Bull and Bear researchers that debate**, a **risk management team**, a synthesising trader. Public repo: github.com/TauricResearch/TradingAgents.

That is this project's design. **"Our contribution is the Bull vs. Bear debate" will not survive a literature review.**

This is *not* fatal. Replication-plus-extension is legitimate and often stronger than a weak novelty claim. But it must be cited and positioned deliberately, in Chapter 2.

5.2 "Proving multi-agent > single LLM" is the wrong goal

The Phase 5 deliverable reads: *"Backtest results **proving** multi-agent > single LLM."*

You cannot set out to prove a conclusion. If the debate layer doesn't help, that is a **result**, not a failure — and a well-measured negative result defends better than a vague positive one.

5.3 Where the real contribution lives

Axis	Why it is defensible
Leakage-free evaluation	Most published multi-agent finance results do not control for it
Debate ablation	Does the debate earn its ~2× token cost? Almost nobody measures this
Confidence calibration	When it says 80%, is it right 80% of the time? Rarely reported
Indian-market focus	The literature is overwhelmingly US-large-cap; NSE is under-studied (see §6)

Proposed research question:

Under an evaluation protocol that provably excludes lookahead bias, does adversarial multi-agent debate improve forecast accuracy and confidence calibration over (a) a single-LLM baseline and (b) the same specialists without debate — and is any gain justified by its token cost?

Answerable either way. That is what makes it a research project.

6. [IN] Indian-Market Free Data Strategy

Scope decision (revised 2026-08-06): NSE/BSE only. The earlier "worldwide coverage" plan is withdrawn. Narrowing is not a retreat — a single market gives one trading calendar, one currency and one regulator, which is what makes the leakage-free *as_of* protocol *provable* rather than approximate. It also buys three India-specific mechanisms (point-in-time universe, SEBI filing lag, NIFTY-relative scoring) that a worldwide build cannot have.

6.1 Price data — yfinance .NS / .BO, with an India-native fallback

Verified 2026-08-06: RELIANCE.NS returns `currency: INR, exchange: NSE, timezone: Asia/Kolkata, history from 1996`.

The Stooq fallback in the previous version of this review is dead. Verified 2026-08-06: `stooq.com/q/d/1/` now answers scripted clients with a **JavaScript proof-of-work challenge page**, not CSV — so the documented fallback would have silently returned nothing on the exact day it was needed. It is replaced by:

	NSE daily bhavcopy archive
URL	<code>nsearchives.nseindia.com/content/cm/BhavCopy_NSE_CM_0_0_0_YYYYMMDD_F_0000.csv.zip</code>
Key	None — but a browser User-Agent is required
Verified	HTTP 200, ~170 KB/day (2025-01-15); 404 on holidays (2025-03-14 Holi) and weekends
Contents	Every NSE cash-market instrument for that day, UDIFF CSV
Caveat	Prices are unadjusted for splits/bonuses — never mix with yfinance auto-adjust rows

It pays for itself three times: **fallback prices, the trading calendar** (file exists NSE traded), and a **survivorship-bias-free universe** (what actually traded on a given date). Cross-check performed: yfinance and bhavcopy agree exactly on RELIANCE 2025-01-15 (1244.95 / 1257.00 / 1241.85 / **1252.20**).

6.2 News — GDELT, scoped to India

Verified: **100+ languages, archive from 1 Jan 1979, updates every 15 minutes, 100% free.** Scope queries with `sourcecountry:india` and the company's registered name.

This solves what NewsAPI cannot: NewsAPI's free tier is **100 req/day, 1-month archive, 24-hour delay, localhost-only** — unusable for any historical evaluation.

[!] Rate-limit trap (verified 2026-08-06): back-to-back GDELT requests return "Please limit requests to one every 5 seconds" as the response body with HTTP 200. It does not raise, and a naive cache will store that sentence as if it were news. Detect the string explicitly, treat it as retryable, and space requests ≥ 5 s.

Trade-offs to document: GDELT gives tone/theme/metadata, **not full article text**. Indian outlets syndicate PTI/ANI copy heavily, so dedup by URL + title similarity is mandatory or polarity skews.

6.3 Macro — FRED's India coverage is thinner than assumed

Verified 2026-08-06 by keyless CSV probe (`fred.stlouisfed.org/graph/fredgraph.csv?id=...`):

Indicator	Series ID	Last observation	Verdict
USD/INR (daily)	DEXINUS	2026-07-31	[OK] Use
India 10Y G-sec yield	INDIRLTLT01STM	2026-05	[OK] Use
India call-money rate	IRSTCI01INM156N	2026-05	[OK] Use (policy-stance proxy)
India CPI	INDCPIALLMINMEI	2025-03	[!] Stale
India IIP	INDPROINDMISMEI	2023-01	[X] Dead
India discount rate	INTDSRINM193N	2022-07	[X] Dead

This would have blocked Phase 2 mid-build. CPI and the RBI repo rate must come from RBI DBIE / MoSPI, or — the honest BTP answer — hardcode the published repo-rate decision calendar and document CPI as a limitation. Do not fabricate a series.

6.4 Recommended stack

Layer	Source	Cost	Status
Price / OHLCV	yfinance .NS/.BO (cached) → NSE bhavcopy fallback	Free	[OK] Verified
Trading calendar	NSE bhavcopy availability	Free	[OK] Verified
Point-in-time universe	NSE bhavcopy daily snapshot	Free	[OK] Verified
Indicators	Computed locally	Free	[OK]
Fundamentals	yfinance [!] <i>not point-in-time, patchier on .NS</i>	Free	Partial
News	GDELT , <code>sourcecountry:india</code>	Free	[OK] (rate-limit handling required)
Macro	FRED (FX, 10Y, call rate) + RBI/MoSPI for CPI & repo	Free	Partial — see §6.3
Benchmark / regime	<code>^NSEI</code> , <code>^BSESN</code> , <code>^INDIAVIX</code> , <code>INR=X</code> , <code>BZ=F</code> , NIFTY sectoral	Free	[OK] Verified

Total data cost: zero.

6.5 The one gap money cannot close

Point-in-time fundamentals. yfinance returns *today's* restated financials, not what was public on your as-of date. True PIT databases (Compustat PIT, FactSet) are institution-priced.

Mitigation, not purchase: apply the **SEBI LODR Reg. 33 filing rule** — a quarter is visible only once `as_of` $\geq$ `quarter_end` + 45 days (annual: 60) — restrict to slow-moving ratios, and **document the limitation explicitly**. The filing rule fixes *timing* leakage; it does not fix *restatement* leakage. A stated limitation earns credit; a hidden one loses it.

7. Security Review

Relevant because this is a project *about* AI agents — examiners will ask.

Risk	Severity	Fix
Prompt injection via headlines — untrusted web text goes straight into the LLM	Medium	Treat headlines as data, not instructions: delimit clearly, instruct the model to ignore embedded directives, validate output against the schema. Discuss in the report — it shows awareness of the risk class.
No auth / rate limit on FastAPI	High <i>if deployed</i>	Per-IP rate limit before any public exposure; otherwise anyone can drain your quota
<code>NEXT_PUBLIC_</code> key leak	Critical	Never prefix secrets with <code>NEXT_PUBLIC_</code> — it ships them into the client bundle. All provider calls stay server-side.
<code>CORS allow_origins=["*"]</code>	Medium	Pin to your frontend origin explicitly
WebSocket has no timeout/connection cap	Low	Idle timeout + max connections
<code>.env</code> committed	Critical	<code>.gitignore</code> exists — verify the key was never committed before pushing
No dependency audit	Low	<code>pip audit</code> before submission

8. [OK] Corrected Phase Plan

The count is not the issue — **the order is**. Build the ruler before the thing it measures.

Phase 1 — Foundation & Walking Skeleton

Environment (fixing §3), **all** API keys including **FRED**, `cache.py` first, `tickers.py` (the India boundary), `market_data.py` with `as_of` + **NSE bhavcopy fallback**, `calendar_nse.py`, indicators, **the single-LLM baseline**, one specialist end-to-end, **and a FinalForecast schema smoke-test against Gemini** (§4.4).

Gate: `python -m forecast <TICKER> --as-of <DATE>` returns a valid schema-conformant forecast twice; the second run makes **zero** network calls. `pip freeze > requirements.txt` committed.

Phase 2 — Evaluation Harness & Leakage-Free Protocol

Determine the training cutoff, run the recall probe, **freeze and commit the test set**, implement metrics (directional accuracy, MAE, **calibration/Brier, cost per forecast**), add a naive baseline, record baseline results.

Gate: any forecaster can be scored by one command, reproducibly; probe result documented.

Phase 3 — Multi-Agent System

Remaining specialists, LangGraph `StateGraph` with **distinct state keys** (§4.5) and checkpointing, **reconciliation gate**, bounded Bull/Bear debate, risk synthesiser, and the `--no-debate` **switch wired from the start**.

Gate: full system scores end-to-end; reconciliation catches a deliberately injected fake number.

Phase 4 — Experiments & Ablation ← *where the marks are*

Multi-agent vs single-LLM vs naive; **debate ablation** (accuracy Δ and cost Δ); specialist leave-one-out; calibration curves; ≥ 3 repeats for variance; qualitative failure analysis on 5–10 wrong forecasts.

Gate: every claim destined for the report traces to a committed results file.

Phase 5 — Dashboard, Report & Defence

FastAPI + **Streamlit first** (React only if time genuinely remains), live agent log, debate view, forecast card with citations. Report with **TradingAgents positioned in Chapter 2**. Slides + **a cached known-good demo run**.

Gate: clean clone runs the demo; every number traces to a results file.

9. [!] What I Could NOT Verify

Listed explicitly so the next reviewer can close them.

Item	Status	Action
Gemini free-tier RPM/RPD	Google's official rate-limit page no longer publishes per-model numbers ; it directs users to AI Studio	Check <code>aistudio.google.com</code> directly before designing around any figure . Treat all quoted numbers — mine or anyone's — as unverified.

Item	Status	Action
Gemini training cutoff date	Not reliably published per model	Determine empirically via the §4.2 recall probe. This is more defensible than citing a date anyway.
Exact yfinance ban threshold	Undocumented by design (unofficial scraper); community reports vary	Assume it is <i>lower</i> than any number you read. Cache-first makes it moot.
TradingAgents' exact datasets/results	Abstract lacks detail	Read the full PDF before writing Chapter 2

Design consequence: because the two most load-bearing numbers (LLM quota, training cutoff) are not reliably published, the architecture must be **quota-adaptive** (backoff + caching + Flash-class models) and the evaluation must be **empirically probed**, not documentation-driven.

10. Bottom Line

Should you start building? Yes — today.

- The architecture is validated by both industry practice and published research.
- The engineering plan is largely sound; most defects here are fix-as-you-go.
- Nothing requires a restart.

But do these three things first, in this order:

1. **Fix the environment** (§3) — ~10 minutes. `import pandas_ta_classic, backend/__init__.py, pip install, pip freeze.`
2. **Add `as_of` to every tool signature** (§4.1) — before writing any Phase 2 body. Cheap now, expensive later.
3. **Fix the evaluation protocol and novelty framing** (§4.2, §5) — ~1 day. This is what stands between a working demo and a defensible thesis.

The honest summary: the previous plan builds a system that *works* but cannot be *evaluated*. The demo would impress; Chapter 5 would not survive questioning. The corrected plan produces both — and the extra cost is roughly one day of planning, paid now instead of a lost results chapter paid later.

References

- TradingAgents: Multi-Agents LLM Financial Trading Framework — **arXiv:2412.20138**
- Detecting Lookahead Bias in LLM Forecasts — **arXiv:2512.23847**
- Summoning the Oracle to Slay It: Mitigating Look-Ahead Bias in Financial Backtesting with LLMs — **arXiv:2605.24564**
- LangGraph — `INVALID_CONCURRENT_GRAPH_UPDATE` error reference, LangChain docs
- Gemini structured output — googleapis/python-genai issue #699; pydantic-ai issue #3483
- pandas-ta-classic — PyPI, v0.6.52
- GDELT Project — gdeltproject.org
- Yahoo Finance exchange & suffix list — help.yahoo.com

Prepared for independent expert review. Findings marked [OK] were verified against primary sources or reproduced locally; items in §9 are explicitly open.